

DEGREES OF FREEDOM ANALYSIS:

1. Draw a flow diagram and choose a system.
2. Choose components.
3. Label the flow diagram to show the components in each stream and any accumulation in the system. Write down any chemical reactions.
4. Count the *number of independent variables*:

Do this by counting	
a. Count the number of independent stream variables.	The number of components in each stream, then adding them all up.
b. Count the number of independent system variables.	The number of independent chemical reactions, plus the number of components that accumulate (or are depleted) in the system.

5. Count the *number of independent equations*:

Comments	
a. Count the number of specified flows.	Usually there is only one flow specified, that serves as the basis.
b. Count the number of stream composition specifications.	If there are N components in a stream, there are at most $N - 1$ specified stream compositions. For a splitter, add $(N - 1)(N_s - 1)$, to the count, where N_s is the number of output streams.
c. Count the number of system performance specifications.	
d. Count the number of material balance equations.	If there are N components in a system, then there are N material balance equations.

6. Calculate $DOF = \text{number of independent variables} - \text{number of independent equations}$.

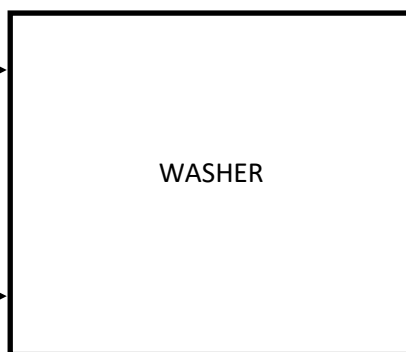
Problem 2.66

1. Raw Pigment Stream: [T, S, W]

$$\begin{aligned}\dot{m}_1 &= 10,000 \frac{\text{lb}}{\text{h}} \\ \omega_{T,1} &= 40 \text{ wt. \%} \\ \omega_{S,1} &= 20 \text{ wt. \%} \\ \omega_{W,1} &= 40 \text{ wt. \%}\end{aligned}$$

2. Washwater Stream: [W]

$$\dot{m}_{W,2} = 6 \frac{\text{lb } W}{\text{lb } T}$$



3. Washed Pigment Stream: [T, S, W]

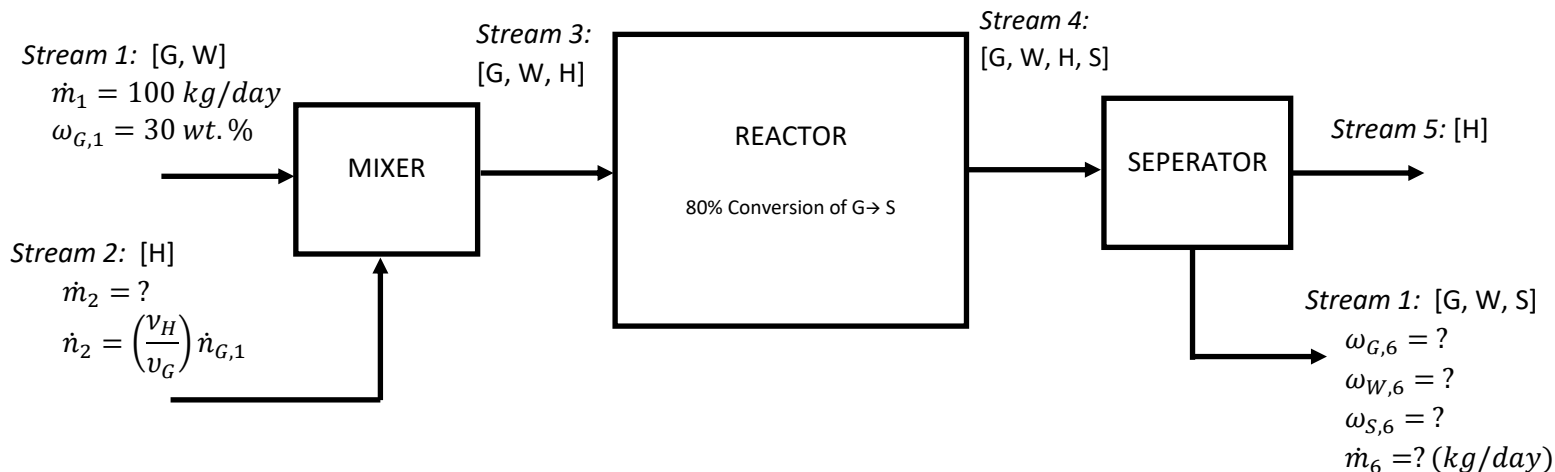
$$\begin{aligned}\dot{m}_{S,4} &= 100 \frac{\text{parts } S}{\text{million parts } T} \\ \omega_{T,3} &= 50 \text{ wt. \%}\end{aligned}$$

4. Wastewater Stream: [W, S]

$$\begin{aligned}\dot{m}_4 &= \max 30,000 \frac{\text{lb}}{\text{h}} \\ \omega_{S,4} &= \max 0.5 \text{ wt. \%}\end{aligned}$$

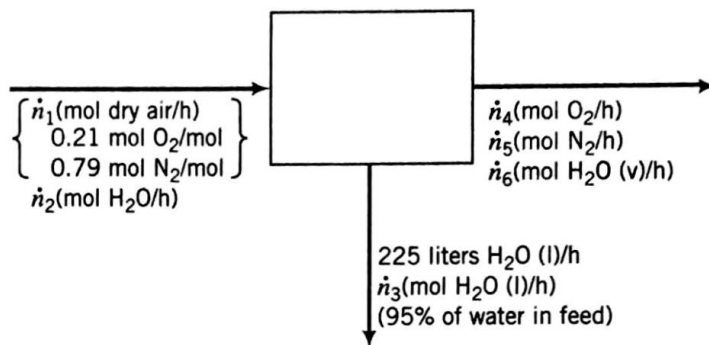
Specified, over specified, or underspecified?

P2.49: Sorbitol ($C_6H_{14}O_6$) is made from glucose ($C_6H_{12}O_6$) and hydrogen. 100kg/day of a 30 wt% glucose solution is mixed with a stoichiometric flow rate of hydrogen and sent to a reactor; 80% of the glucose is converted to sorbitol. The hydrogen is then separated from the sugar solution as a gas stream. How much hydrogen (kg/day) is fed to the process? What is the composition (wt%) and flow rate (kg/day) of the liquid stream leaving the process?



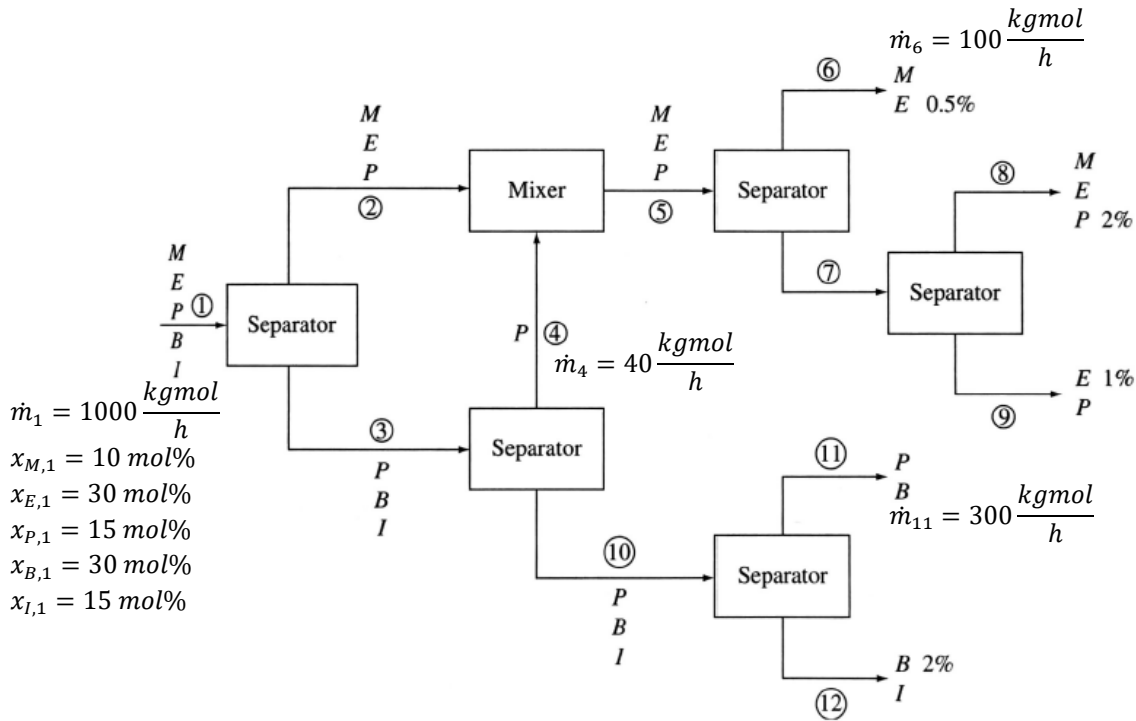
Specified, over specified, or underspecified?

Example 3: A stream of humid air enters a condenser in which 95% of the water vapor in the air is condensed. The flow rate of the condensate (the liquid leaving the condenser) is measured and found to be 225 L/h. Dry air may be taken to contain 21 mole% oxygen, with the balance nitrogen. Calculate the flow rate of the gas stream leaving the condenser and the mole fractions of oxygen, nitrogen, and water stream.



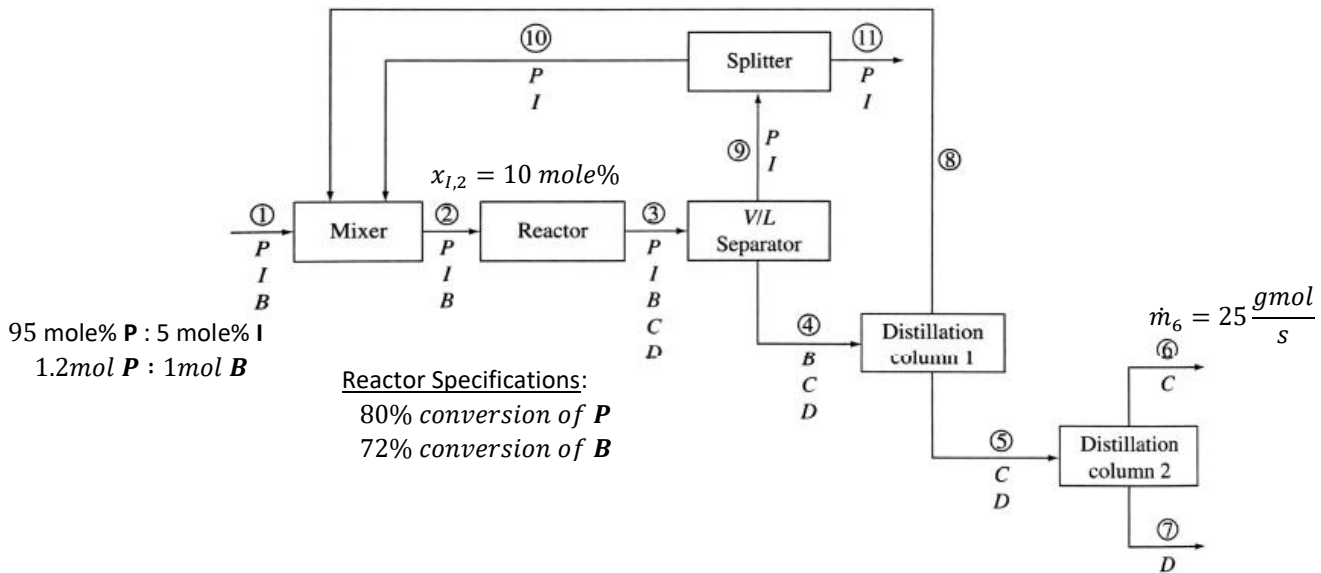
Specified, over specified, or underspecified?

P2.70



Specified, over specified, underspecified?

P2.69



Specified, over specified, underspecified?

MATERIAL BALANCES:***Combustion of a hydrocarbon fuel of unknown composition***

A hydrocarbon gas is burned with air. The dry-basis product gas composition is 1.5 mole% CO, 6.0% CO₂, 8.2% O₂, and 84.3% N₂. There is no atomic oxygen in the fuel. Calculate the ratio of hydrogen to carbon in the fuel gas and speculate on what the fuel might be.